

Yonghoon Jeong

Republic of Korea Naval Academy

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Research Interests

Wireless Communication Systems
Performance Analysis and Modeling
Physical-Layer Wireless Sensing and Networking
Low-Power Wide-Area Networks (LPWANs)
Ocean Internet of Things (Ocean IoT)

Education

M.S. in Electrical and Computer Engineering | 2024 - 2026

- Seoul National University, Seoul, Republic of Korea
- Advisor: Prof. Saewoong Bahk (currently President of ETRI)

B.S. in Electronics and Communications Engineering | 2018 - 2023

- Kwangwoon University, Seoul, Republic of Korea

Professional Experience

Republic of Korea Naval Academy | 2026 - Present

- Faculty Member, Department of Cyber Science
- Researcher, Institute of Naval Research
- Commissioned as a Republic of Korea Navy officer.
- Plan to pursue a Ph.D. abroad following military service completion in May 2029.

Seoul National University (SNU) | 2024 - 2026

- Researcher, Network Laboratory (NETLAB), Institute of New Media and Communications (INMC)

Electronics and Telecommunications Research Institute (ETRI) | 2023

- Researcher, AI-SoC Infrastructure Research Section

Korea Advanced Institute of Science and Technology (KAIST) | 2022

- Research Intern, Unmanned Systems Research Group, School of Electrical Engineering
- Advisor: Prof. David Hyunchul Shim

Publications

International Papers

- "On the Symbol Error Rate of Underwater Acoustic Backscatter Systems." **Under review** at *IEEE Transactions on Communications*.
- "Underwater Backscatter for Ocean Sensing: From Wireless Power to Data Collection." **Under review** at *IEEE Wireless Communications* (Magazine).
- "Characterizing the Relation Between the Energy Scattering Effect and Symbol Error Rate in Non-Linear Chirp Modulation for LoRa." **Under review** at *IEEE Wireless Communications Letters*.
- "Symbol Error Rate Analysis and Approximation for Arbitrary Chirp Modulation Under Multipath, Doppler, and Interference." **Under review** at *IEEE Transactions on Communications*.
- "On the Symbol Error Rate of Non-Linear Chirp Modulation for LoRa Under Asynchronous Interference." **Under review** at *IEEE Wireless Communications Letters*.

- **Yonghoon Jeong**, Youngwook Son, and Saewoong Bahk, “On the Interference Resilience of Non-Linear Chirp Modulation for LoRa: An Analytical Insight into Energy Scattering Effect.” **Published** in *IEEE Communications Letters*, vol. 30, pp. 2238-2242, Jun. 2026. **(Paper)**
- Dongrak Choi, Yubin Choi, **Yonghoon Jeong**, Jeongyeup Paek, and Saewoong Bahk, “D²-PLC: Holistic Design and Implementation of High-Data-Rate Duplex DC Power Line Communication Network.” **Published** in *IEEE Internet of Things Journal*, vol. 12, pp. 45325-45340, Nov. 2025. **(Paper)**
- Dongrak Choi, **Yonghoon Jeong**, Yubin Choi, and Saewoong Bahk, “Demo: A Programmable High-Throughput Duplex DC-PLC Testbed for Power and Data Integration.” **Published** in the proceedings of *IEEE ICNP 2025*, Seoul, Republic of Korea, Sep. 22-25, 2025. **(Paper)**
- **Yonghoon Jeong** and Youngseok Choi, “Diffusion-Phys: Noise-Robust Heart Rate Estimation from Facial Videos via Diffusion Models.” **Published** in *Biomedical Engineering Letters*, vol. 15, pp. 575-585, Apr. 2025. **(Paper)**

Domestic Papers

- **Yonghoon Jeong** and Saewoong Bahk, “A Study on Direct Current Power Line Communication (DC-PLC) with Applications in Automotive, Maritime, and Aviation.” **Published** in the proceedings of *The Conference of the Korea Institute of Communications and Information Sciences (KICS)*, Gyeongju, Republic of Korea, Nov. 20-22, 2024.
- **Yonghoon Jeong**, “Development of Non-contact Heart Rate Measurement Technology Based on Facial Video.” **Published** in the proceedings of *The Conference of the Korean Institute of Information Scientists and Engineers (KIISE)*, Jeju, Republic of Korea, Jun. 23-25, 2021.

Academic Service

Journal Reviewer: IEEE Internet of Things Journal

Awards & Scholarships

Scholarship Recipient (2024 - 2025) - Brain Korea 21 (BK21), National Research Foundation of Korea
Scholarship Recipient (2022 - 2023) - Korea Scholarship Foundation for the Future Leaders (KOSFFL)
Excellent Paper Award (2021) - Korea Computer Conference 2021
University President's Award (2020) - Seoul Capstone Design, Seoul
Scholarship Recipient (2020) - Samsung Scholarship Foundation
Dean's List (2020) - Highest GPA

Teaching Experience

Creative Engineering Design (2024) - Teaching Assistant; Prof. Saewoong Bahk
Electronic Circuits Experiment I (2023) - Instructor; Emeritus Prof. Young-chul Chung
Electronic Circuits Experiment II (2023) - Instructor; Emeritus Prof. Young-chul Chung
Electronic Circuits Experiment I (2022) - Instructor; Prof. Youngseok Choi

Technical Skills

Programming Languages: C, C++, Python, MATLAB
Frameworks & Libraries: ROS, ROS2, TensorFlow, PyTorch
Simulation Tools: MATLAB/Simulink, ns-3
Development Tools: Linux, Git, Docker